

HW1

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Section 1.3 #27

Find all 2×2 matrices of the form

$$A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$$

such that

$$A^2 = AA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

solving A^2 ,

$$A^2 = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} = \begin{pmatrix} a^2 & b(a+c) \\ 0 & c^2 \end{pmatrix}.$$

set this answer equal to I

$$a^2 = 1, \quad c^2 = 1, \quad b(a+c) = 0.$$

the first two equations give us $a = \pm 1$ and $c = \pm 1$. and for the third equation, if $a = c$ then $a + c \neq 0$, so $b = 0$; if $a \neq c$ then $a + c = 0$ automatically, so b has got to be any real number. this gives all solutions:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & b \\ 0 & -1 \end{pmatrix}, \quad b \in \mathbb{R}, \quad A = \begin{pmatrix} -1 & b \\ 0 & 1 \end{pmatrix}, \quad b \in \mathbb{R}.$$

Section 1.4 #5

Section 1.5 #17

Section 1.5 #29